

- 4 Boards used for chopping fruit and vegetables can be made from wood or oil-based plastics.

The photograph shows some chopping boards.



(Source: © Dorling Kindersley Ltd/Alamy Stock Photo)

- (a) (i) Bacteria can grow on the surfaces of chopping boards.

Explain the conditions that would result in the greatest bacterial growth on the surface of a chopping board.

(3)

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- (ii) Explain why boards made from oil-based plastics are not considered sustainable.

(2)

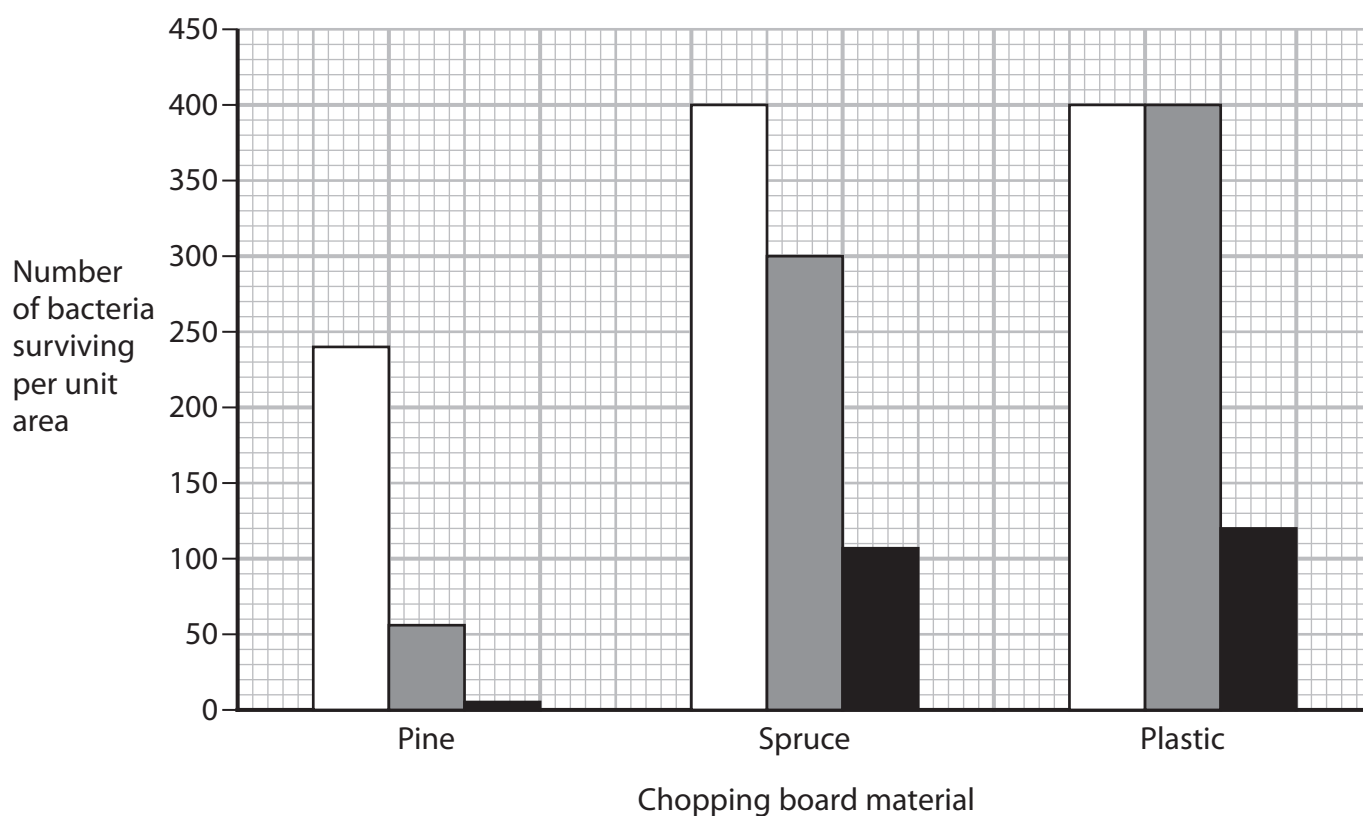
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- (b) The number of bacteria surviving on boards made from different materials was investigated.

Identical bacterial cultures were placed on boards made of pine wood, spruce wood and plastic. The graph shows the number of bacteria surviving after bacteria were placed on the boards and kept in the same conditions.



Key

1 hour 4 hours 24 hours



(i) Comment on the results of this investigation.

(3)

(ii) Suggest **one** reason for the difference in bacterial survival on the pine chopping board and on the plastic chopping board.

(1)

(Total for Question 4 = 9 marks)

